SALES DATA ANALYSIS REPORT

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Task: Sales Data Analysis
Date: 8 October 2026

1. OBJECTIVE

To analyze the 2024 sales data and extract meaningful business insights including top products, regions, categories, payment trends, and monthly patterns.

2. TOOLS USED

- Python 3.14.7
- pandas (data loading, groupby, aggregation)
- matplotlib (visualization)
- Microsoft Excel (data cleaning)

3. DATASET

- Total Orders: 55
- Total Columns: 14
- Time Period: January 2024 – November 2024
- Categories: Electronics, Furniture, Clothing
- Regions: North, South, East, West
- Payment Methods: UPI, Credit Card, Debit Card, Cash

4. METHODOLOGY

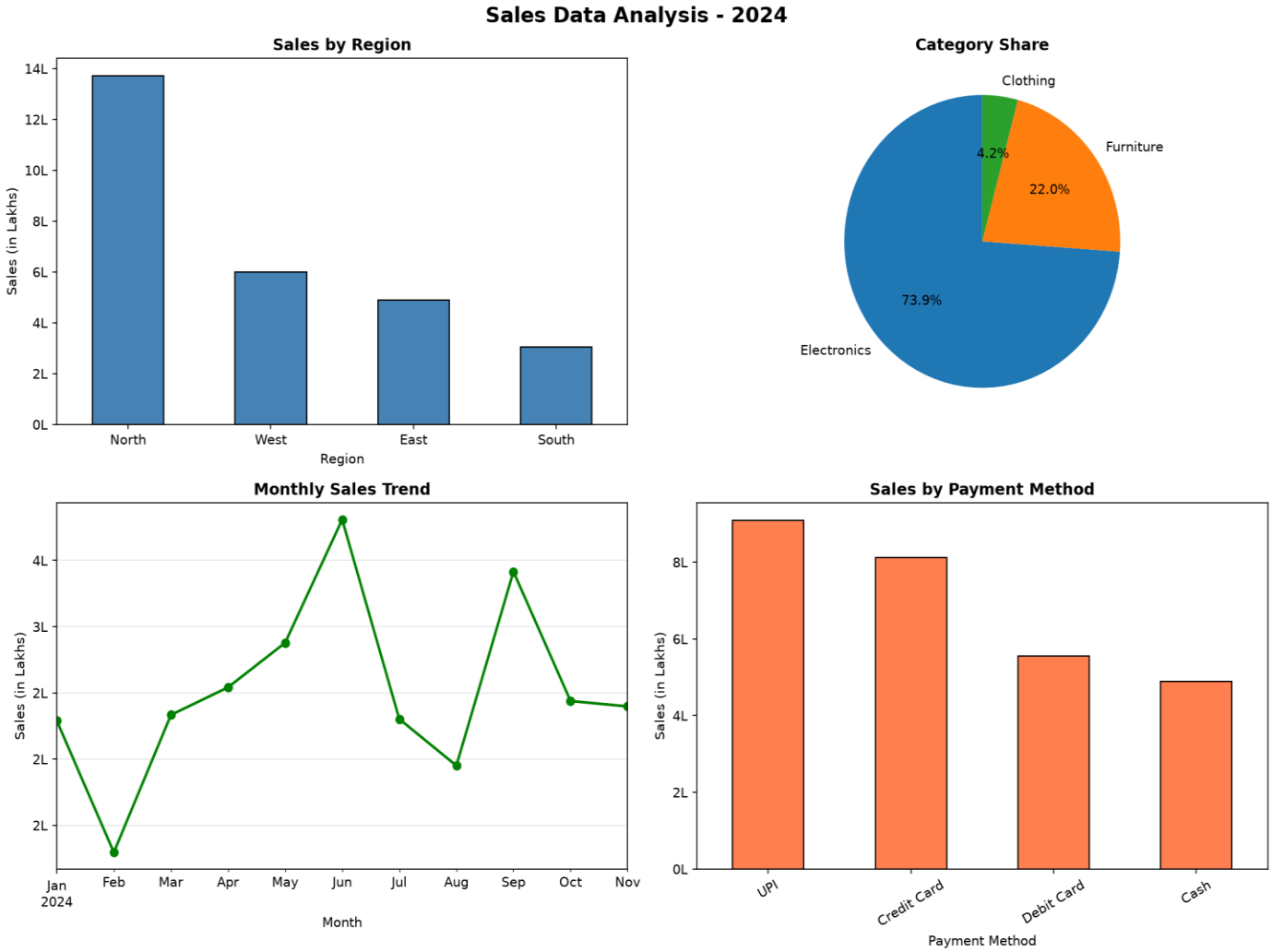
Step 1: Data was cleaned in Excel (missing values, duplicates, date formatting issues)
Step 2: The cleaned dataset was loaded into pandas using pd.read_excel()
Step 3: Key performance indicators were calculated using groupby() and aggregation functions
Step 4: Four charts were created using matplotlib for visualization

5. KEY FINDINGS

- Total Sales: ₹27,65,230
- Total Orders: 55
- Total Quantity Sold: 287
- Average Order Value: ₹50,277
- Top Region: North
- Top Category: Electronics (73.9% share)

- Top Payment Method: UPI
- Top Product: TV
- Top Salesperson: Amit
- Top Customer: Rahul Sharma

6. VISUALIZATIONS



- Chart 1: Sales by Region (Bar Chart)
- Chart 2: Category Share (Pie Chart)
- Chart 3: Monthly Sales Trend (Line Chart)
- Chart 4: Sales by Payment Method (Bar Chart)

7. INSIGHTS

- The North region is the most profitable, driven by high-value electronics products
- The Electronics category contributes 73.9% of total revenue.
- UPI is the most preferred payment method among customers.
- June 2024 was the peak month due to high-value orders such as cameras and laptops.

- February 2024 recorded the lowest sales due to fewer orders and low-value products.
- Five orders were cancelled, resulting in an approximate 8% revenue loss.

8. RECOMMENDATIONS

- Increase Electronics inventory in the North region.
- Promote UPI as a payment method to reduce transaction costs.
- Investigate the reasons behind cancelled orders.
- Launch targeted marketing campaigns during low-performing months (February, August).
- Document the strategies used by the top salesperson, Amit.

9. CONCLUSION

The Electronics category and North region are the most profitable segments for the business. UPI payments dominate customer transactions. Reducing cancelled orders could potentially increase revenue by approximately 8%.

10. FILES SUBMITTED

sales_analysis.py —> Python script
sales_analysis_charts.png —> Visualizations
sales_data_cleaned.xlsx —> Cleaned dataset
Sales_Analysis_Report.pdf —> This report